



Engineering Electromagnetics

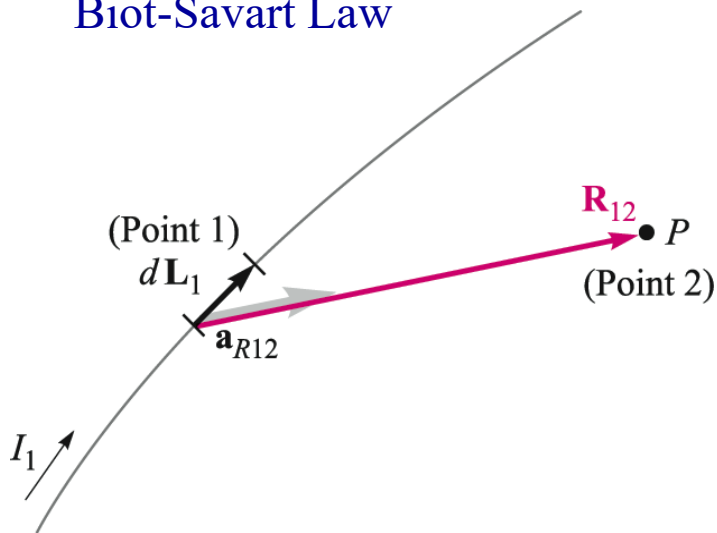
工程电磁场

Youwei Jia 嘉有为

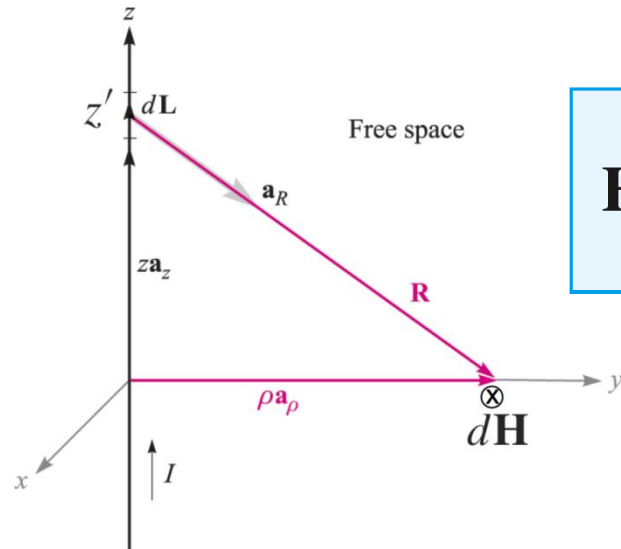
11: The Steady Magnetic Field

- Review

Biot-Savart Law

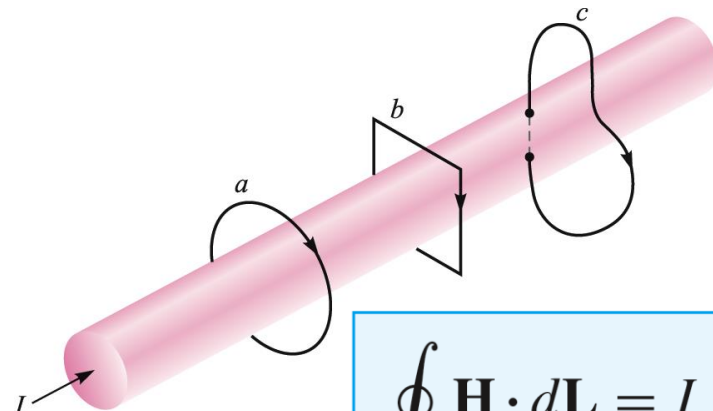


$$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$$



$$\mathbf{H}_2 = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

Ampere's Circuital Law

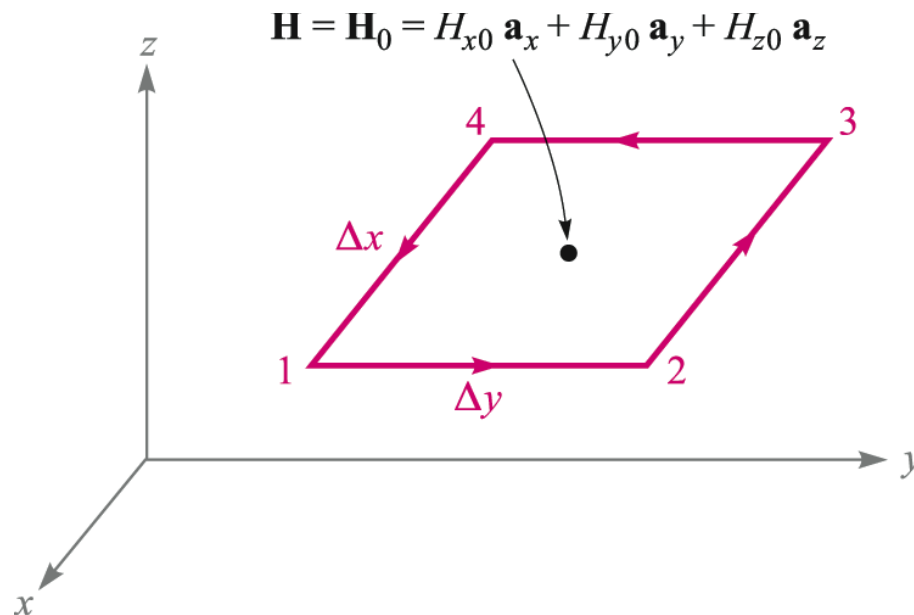


$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

11: The Steady Magnetic Field

- Ampere's Law as Applied to a Small Closed Loop

Consider magnetic field $\mathbf{H}$ evaluated at the point shown in the figure. We can approximate the field over the closed path 1234 by making appropriate adjustments in the value of $\mathbf{H}$ along each segment.



The objective is to take the closed path integral and ultimately obtain the point form of Ampere's Law 安培定律的微分形式.

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- Ampere's Law as Applied to a Small Closed Loop

Approximation of $\mathbf{H}$ Along One Segment

Along path 1-2, we may write:

$$(\mathbf{H} \cdot \Delta \mathbf{L})_{1-2} = H_{y,1-2} \Delta y$$

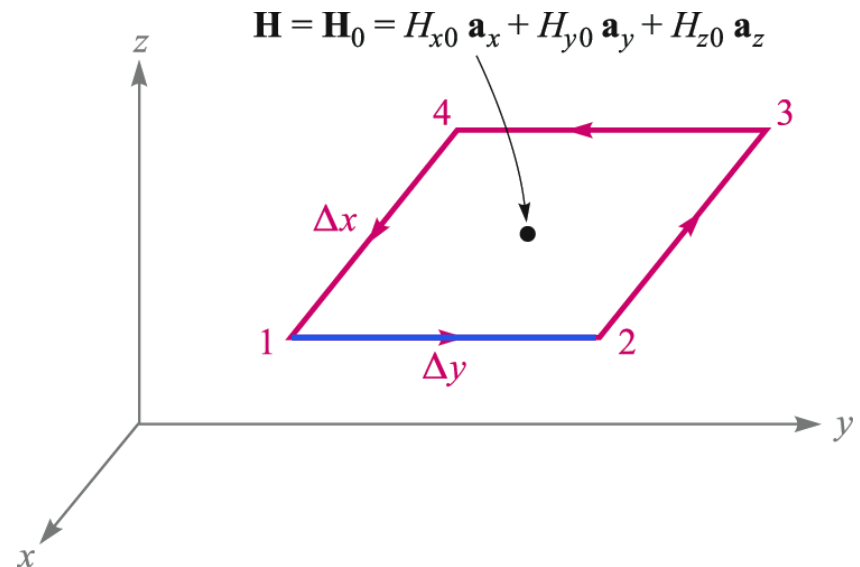
where:

$$H_{y,1-2} \doteq H_{y0} + \frac{\partial H_y}{\partial x} \left(\frac{1}{2} \Delta x \right)$$

Note: Please refer to Taylor series expansion

And therefore:

$$(\mathbf{H} \cdot \Delta \mathbf{L})_{1-2} \doteq \left(H_{y0} + \frac{1}{2} \frac{\partial H_y}{\partial x} \Delta x \right) \Delta y$$



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- Ampere's Law as Applied to a Small Closed Loop

Contributions of y-Directed Path Segments

The contributions from the front and back sides will be:

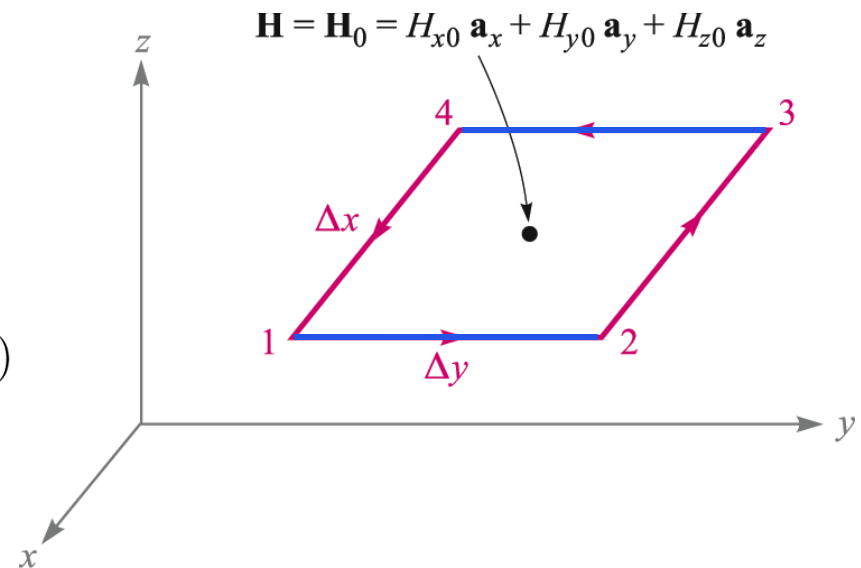
$$(\mathbf{H} \cdot \Delta \mathbf{L})_{1-2} \doteq \left(H_{y0} + \frac{1}{2} \frac{\partial H_y}{\partial x} \Delta x \right) (\Delta y)$$

The contribution from the opposite side is:

$$(\mathbf{H} \cdot \Delta \mathbf{L})_{3-4} \doteq \left(H_{y0} - \frac{1}{2} \frac{\partial H_y}{\partial x} \Delta x \right) (-\Delta y)$$

Note the **path directions** as specified in the figure, and how these determine the **signs** used.

This leaves the left and right sides.....



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- Ampere's Law as Applied to a Small Closed Loop

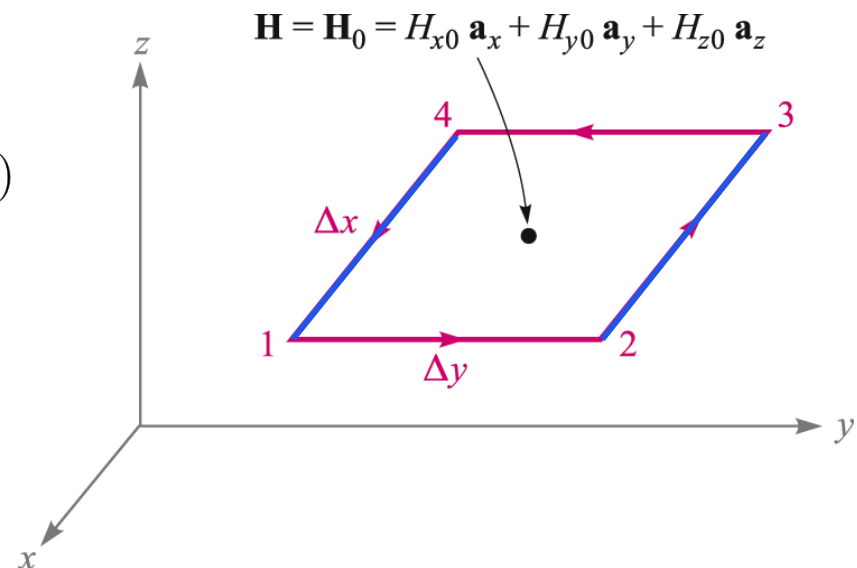
Contributions of x -Directed Path Segments

Along the right side (path 2-3):

$$(\mathbf{H} \cdot \Delta \mathbf{L})_{2-3} \doteq \left(H_{x0} + \frac{1}{2} \frac{\partial H_x}{\partial y} \Delta y \right) (-\Delta x)$$

...and the contribution from the left side (path 4-1) is:

$$(\mathbf{H} \cdot \Delta \mathbf{L})_{4-1} \doteq \left(H_{x0} - \frac{1}{2} \frac{\partial H_x}{\partial y} \Delta y \right) (\Delta x)$$



The next step is to add the contributions of all four sides to find the closed path integral:

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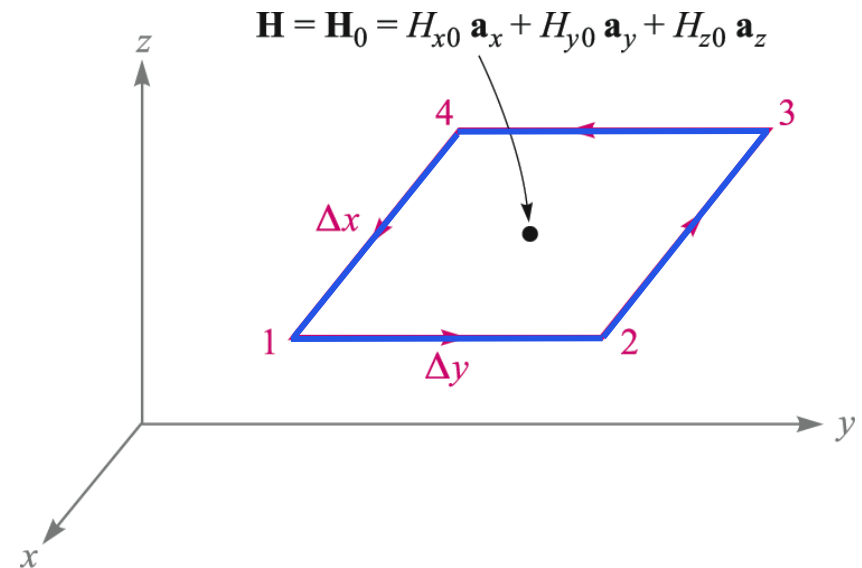
- Ampere's Law as Applied to a Small Closed Loop

The total integral will now be the sum:

$$\oint \mathbf{H} \cdot d\mathbf{L} \doteq (\mathbf{H} \cdot \Delta\mathbf{L})_{1-2} + (\mathbf{H} \cdot \Delta\mathbf{L})_{2-3} + (\mathbf{H} \cdot \Delta\mathbf{L})_{3-4} + (\mathbf{H} \cdot \Delta\mathbf{L})_{4-1}$$

and using our previous results, the becomes:

$$\oint \mathbf{H} \cdot d\mathbf{L} \doteq \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \Delta x \Delta y$$



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• Ampere's Law as Applied to a Small Closed Loop

By Ampere's Law, the closed path integral of $\mathbf{H}$ is equal to the enclosed current, approximated in this case by the current density at the center, multiplied by the loop area:

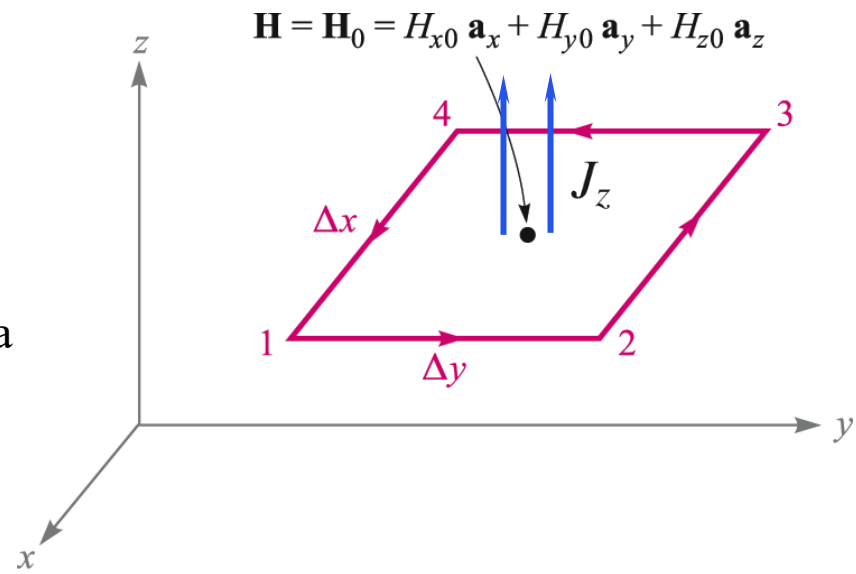
$$\oint \mathbf{H} \cdot d\mathbf{L} \doteq \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \Delta x \Delta y \doteq J_z \Delta x \Delta y$$

Dividing by the loop area, we now have:

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta x \Delta y} \doteq \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \doteq J_z$$

The expression becomes exact as the loop area approaches zero:

$$\lim_{\Delta x, \Delta y \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta x \Delta y} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = J_z$$



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- Ampere's Law as Applied to a Small Closed Loop

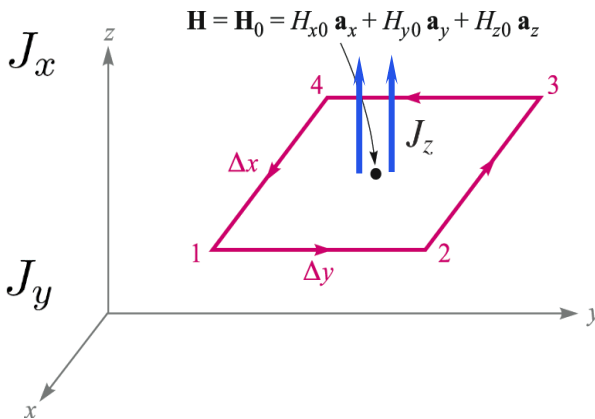
Other Loop Orientations

The same exercise can be carried with the rectangular loop in the other two orthogonal orientations. The results are:

Loop in yz plane: $\lim_{\Delta y, \Delta z \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta y \Delta z} = \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = J_x$

Loop in xz plane: $\lim_{\Delta z, \Delta x \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta z \Delta x} = \frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = J_y$

Loop in xy plane: $\lim_{\Delta x, \Delta y \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta x \Delta y} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = J_z$



This gives all three components of the current density field.



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- Curl of a Vector Field 矢量场的旋度, 旋度场

The previous exercise resulted in the rectangular coordinate representation of the Curl of $\mathbf{H}$.

In general, the curl of a vector field is another field that is normal to the original field. The curl component in the direction N , normal to the plane of the integration loop is:

$$(\text{curl } \mathbf{H})_N = \lim_{\Delta S_N \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta S_N}$$

where ΔS_N is the planar area enclosed by the closed line integral.

The direction of N is taken using the right-hand convention: With fingers of the right hand oriented in the direction of the path integral, the thumb points in the direction of the normal (or curl).

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- Curl in Rectangular Coordinates

Assembling the results of the rectangular loop integration exercise, we find the vector field that comprises curl $\mathbf{H}$:

$$\text{curl } \mathbf{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \mathbf{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \mathbf{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \mathbf{a}_z$$

An easy way to calculate this is to evaluate the following determinant:

$$\text{curl } \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

which we see is equivalent to the cross product of the **del operator** with the field:

$$\text{curl } \mathbf{H} = \nabla \times \mathbf{H}$$

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- Curl in Other Coordinate Systems

...a little more complicated!

$$\begin{aligned}\nabla \times \mathbf{H} = & \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \mathbf{a}_\phi \\ & + \left(\frac{1}{\rho} \frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{1}{\rho} \frac{\partial H_\rho}{\partial \phi} \right) \mathbf{a}_z \quad (\text{cylindrical})\end{aligned}$$

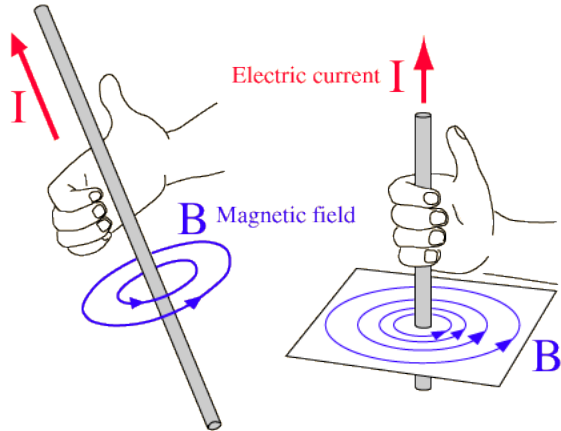
$$\begin{aligned}\nabla \times \mathbf{H} = & \frac{1}{r \sin \theta} \left(\frac{\partial(H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \mathbf{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial(r H_\phi)}{\partial r} \right) \mathbf{a}_\theta \\ & + \frac{1}{r} \left(\frac{\partial(r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \mathbf{a}_\phi \quad (\text{spherical})\end{aligned}$$

Look these up as needed....

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- Visualization of Curl





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- Another Maxwell Equation: Point form of Ampere's Law

It has just been demonstrated that:

$$\begin{aligned}\text{curl } \mathbf{H} = \nabla \times \mathbf{H} = & \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \mathbf{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \mathbf{a}_y \\ & + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \mathbf{a}_z = \mathbf{J}\end{aligned}$$

.....which is in fact one of **Maxwell's** equations for static fields:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

This is **Ampere's** Circuital Law in point form.



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- Curl of Conservative Field

$$(\text{curl } \mathbf{H})_N = \lim_{\Delta S_N \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta S_N}$$

We already know that for a static electric field:

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

This means that: $\nabla \times \mathbf{E} = 0$ (applies to a static electric field)

Recall the condition for a conservative field: that is, its closed path integral is zero everywhere.

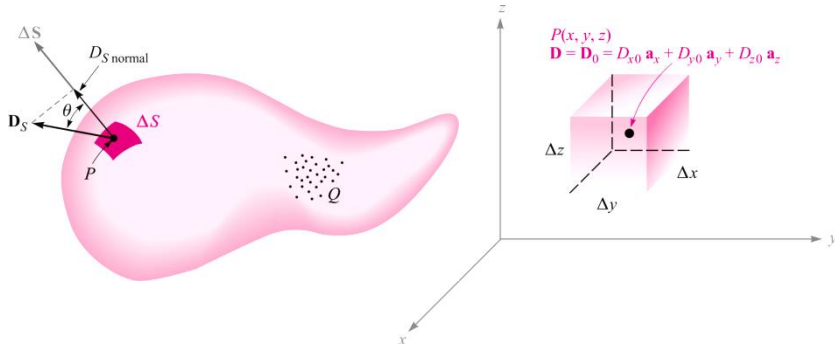
Therefore, a field is conservative if it has zero curl at all points over which the field is defined.

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• Divergence vs. Curl 散度, 旋度

Gauss' Law

$$Q = \oint_S \mathbf{D}_S \cdot d\mathbf{S}$$



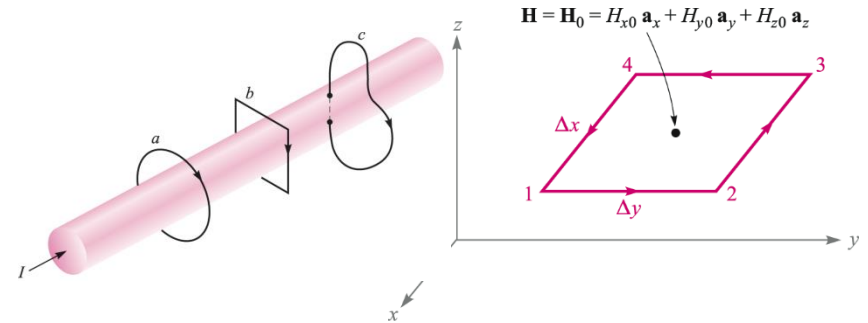
$$\text{Divergence of } \mathbf{A} = \text{div } \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

$$\text{div } \mathbf{D} = \rho_v \quad \text{div } \mathbf{D} = \nabla \cdot \mathbf{D} = \rho_v$$

$$\nabla \cdot \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \text{div}(\mathbf{D})$$

Ampere's Law

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$



$$(\text{curl } \mathbf{H})_N = \lim_{\Delta S_N \rightarrow 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{\Delta S_N}$$

$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\text{curl } \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

11: The Steady Magnetic Field

- Curl Applied to Partitions of a Large Surface

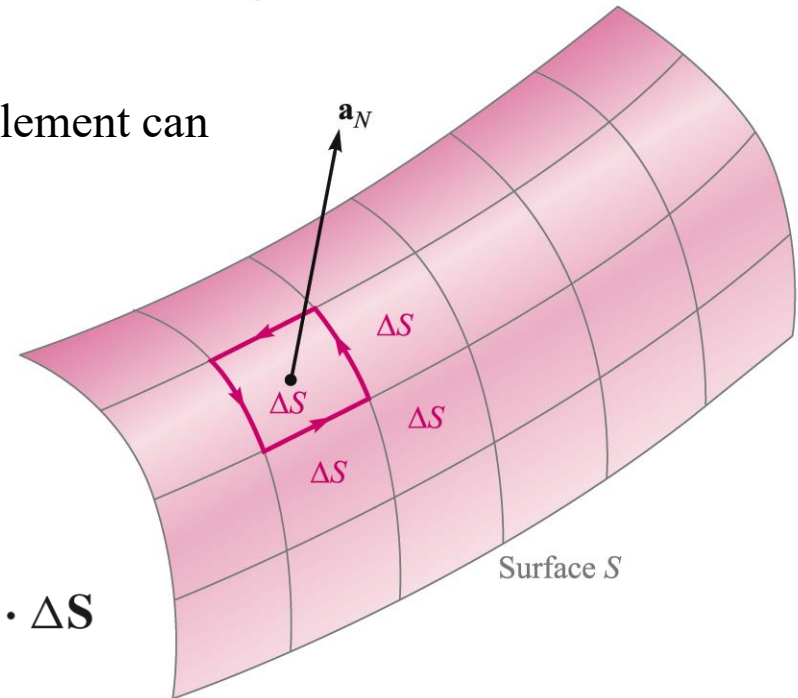
Surface S is partitioned into sub-regions, each of small area ΔS

The curl component that is normal to a surface element can be written using the definition of curl:

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N$$

or:

$$\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N \Delta S = (\nabla \times \mathbf{H}) \cdot \Delta \mathbf{S}$$



We now apply this to every partition on the surface, and add the results....

11: The Steady Magnetic Field

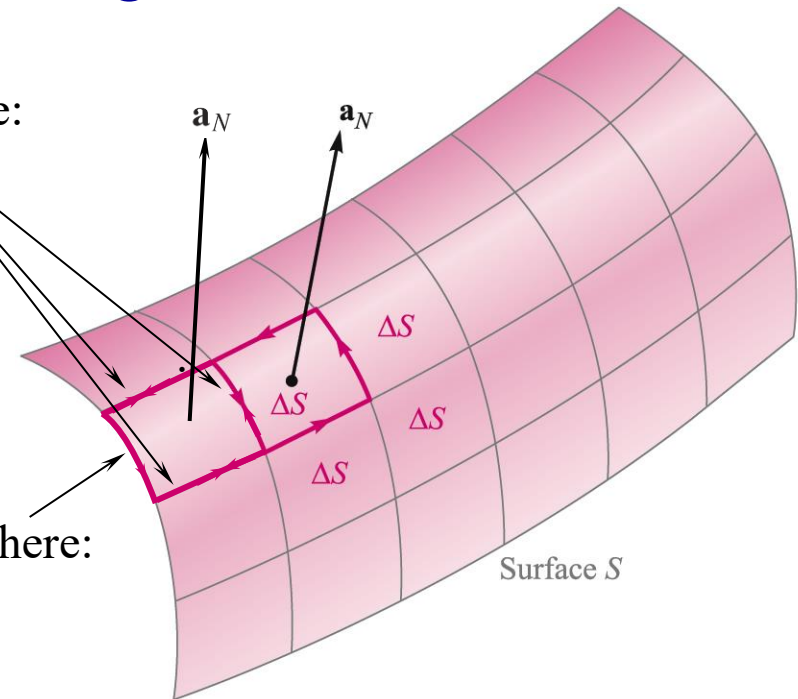
- Curl Applied to Partitions of a Large Surface

We now evaluate and add the curl contributions from all surface elements, and note that adjacent path integrals will all cancel!

This means that **the only contribution to the overall path integral will be around the outer periphery of surface S .**

Cancellation here:

No cancellation here:



Using our previous result, we now write:

$$\sum_{\text{all surface elements}} \oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S} \doteq \sum_{\text{all surface elements}} \nabla \times \mathbf{H} \cdot \mathbf{a}_N \Delta S$$

11: The Steady Magnetic Field

- Stokes' Theorem

We now take our previous result, and take $\Delta S \rightarrow 0$: as

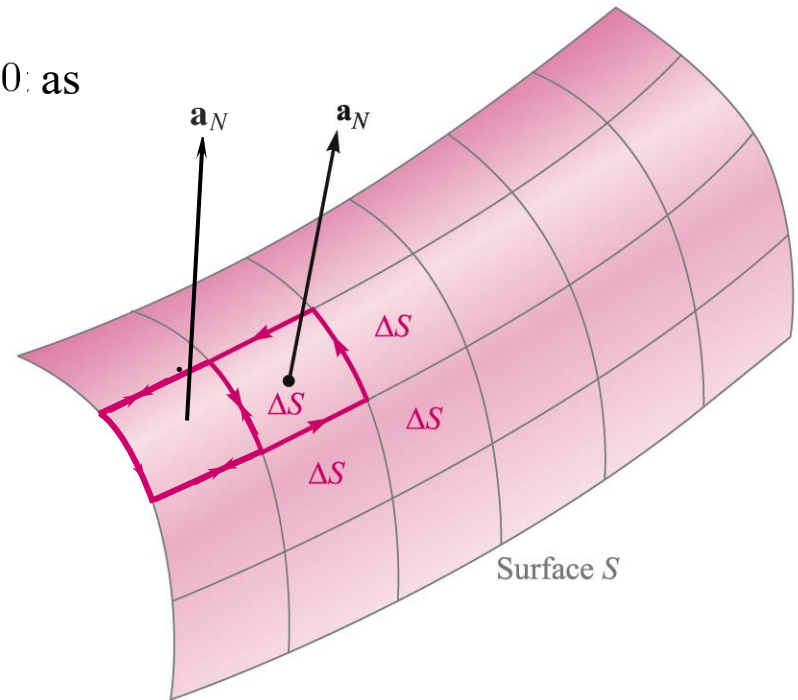
$$\underbrace{\sum_{\text{all surface elements}} \oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}_{\text{In the limit, this side becomes the path integral of } \mathbf{H} \text{ over the outer perimeter because all interior paths cancel}} \doteq \underbrace{\sum_{\text{all surface elements}} \nabla \times \mathbf{H} \cdot \mathbf{a}_N \Delta S}_{\text{In the limit, this side becomes the integral of the curl of } \mathbf{H} \text{ over surface } S}$$

In the limit, this side becomes the path integral of $\mathbf{H}$ over the outer perimeter because all interior paths cancel

In the limit, this side becomes the integral of the curl of $\mathbf{H}$ over surface S

The result is **Stokes' Theorem**

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$



This is a valuable tool to have at our disposal, because it gives us two ways to evaluate the same thing!

11: The Steady Magnetic Field

- Obtaining Ampere's Circuital Law in Integral Form, using Stokes' Theorem

Begin with the point form of Ampere's Law for static fields:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

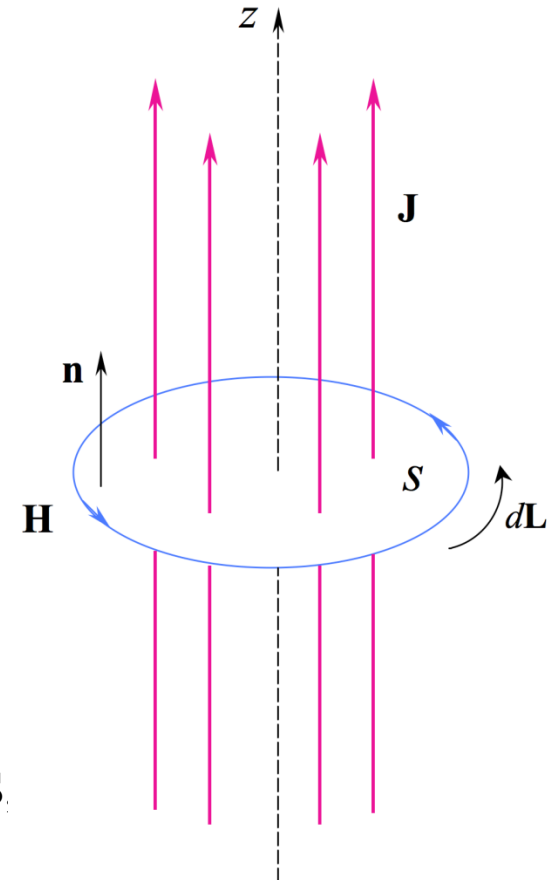
Integrate both sides over surface S :

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \int_S \mathbf{J} \cdot d\mathbf{S} = \oint \mathbf{H} \cdot d\mathbf{L}$$

..in which the far right hand side is found from the left hand side using Stokes' Theorem. The closed path integral is taken around the perimeter of S . Again, note that we use the right-hand convention in choosing the direction of the path integral.

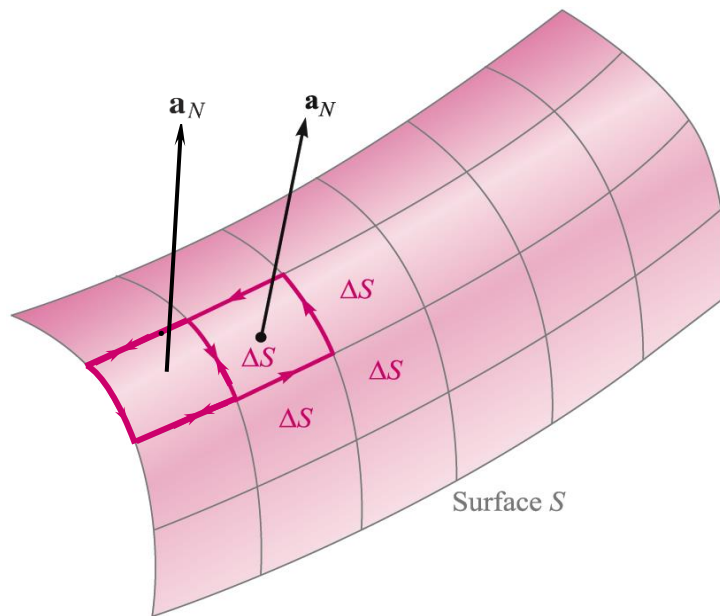
The center expression is just the net current through surface S , so we are left with the integral form of Ampere's Law:

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

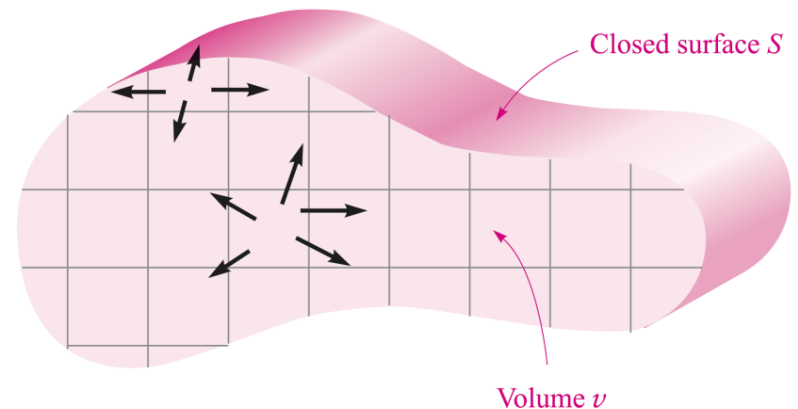


11: The Steady Magnetic Field

- Stokes' Theorem vs. Divergence Theorem



$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$



$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D} dv$$

Note: $\oint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = 0$

$$\nabla \cdot \nabla \times \mathbf{A} \equiv 0$$



11: The Steady Magnetic Field

- Magnetic Flux and Magnetic Flux Density

We are already familiar with the concept of **electric flux**:

$$\Psi = \int_s \mathbf{D} \cdot d\mathbf{S} \quad \text{Coulombs}$$

in which the electric flux density **in free space** is: $\mathbf{D} = \epsilon_0 \mathbf{E}$ C/m²

and where the free space permittivity is $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

In a similar way, we can define the **magnetic flux in units of Webers (Wb)**:

$$\Phi = \int_s \mathbf{B} \cdot d\mathbf{S} \quad \text{Webers}$$

in which the **magnetic flux density** (or **magnetic induction** 磁感应强度) in free space is:

$$\mathbf{B} = \mu_0 \mathbf{H} \quad \text{Wb/m}^2$$

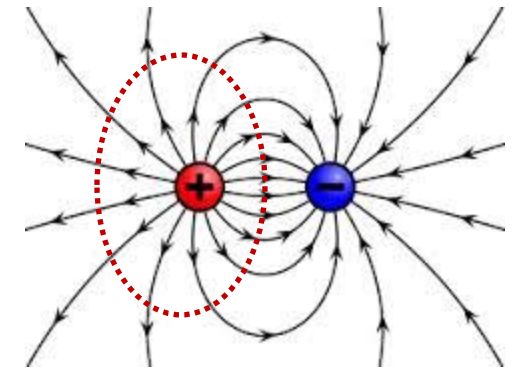
and where **the free space permeability** 磁导率 is $\mu_0 = 4\pi \times 10^{-7}$ H/m

11: The Steady Magnetic Field

- A Key Property of **B**

If the flux is evaluated **through a closed surface**, we have in the case of electric flux, Gauss' Law:

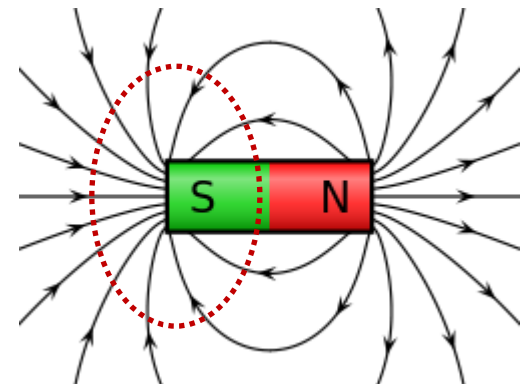
$$\Psi_{net} = \oint_s \mathbf{D} \cdot d\mathbf{S} = Q_{enc}$$



If the same were to be done with magnetic flux density, we would find:

$$\Phi_{net} = \oint_s \mathbf{B} \cdot d\mathbf{S} = 0$$

The implication is that (for our purposes) **there are no magnetic charges** -- specifically, **no point sources of magnetic field exist**. A hint of this has already been observed, in that **magnetic field lines always close on themselves**.





11: The Steady Magnetic Field

- Another Maxwell Equation

We may rewrite the closed surface integral of $\mathbf{B}$ using the divergence theorem, in which the right hand integral is taken over the volume surrounded by the closed surface:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{B} dv = 0$$

Because the result is zero, it follows that

$$\nabla \cdot \mathbf{B} = 0$$

This result is known as Gauss' Law for the magnetic field in point form.



11: The Steady Magnetic Field

- Maxwell's Equations for Static Fields

We have now completed the derivation of Maxwell's equations for no time variation. In point form, these are:

$$\nabla \cdot \mathbf{D} = \rho_v$$

Gauss' Law for the electric field

$$\nabla \times \mathbf{E} = 0$$

Conservative property of the static electric field

$$\nabla \times \mathbf{H} = \mathbf{J}$$

Ampere's Circuital Law

$$\nabla \cdot \mathbf{B} = 0$$

Gauss' Law for the Magnetic Field

where, in free space:

$$\mathbf{D} = \epsilon_0 \mathbf{E}$$

$$\mathbf{B} = \mu_0 \mathbf{H}$$

Note: Significant changes in the above four equations will occur when the fields are allowed to vary with time, as we will see later.



11: The Steady Magnetic Field

- Maxwell's Equations for Static Fields in Large Scale Form

The **divergence theorem** and **Stokes' theorem** can be applied to the previous four point form equations to yield the integral form of Maxwell's equations for static fields:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dV$$

Gauss' Law for the electric field

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

Conservative property of the static electric field

$$\oint \mathbf{H} \cdot d\mathbf{L} = I = \int_S \mathbf{J} \cdot d\mathbf{S}$$

Ampere's Circuital Law

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Gauss' Law for the magnetic field

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- Example: Magnetic Flux Within a Coaxial Line

Consider a length d of coax, as shown here. The magnetic field strength between conductors is:

$$H_{\phi} = \frac{I}{2\pi\rho} \quad (a < \rho < b)$$

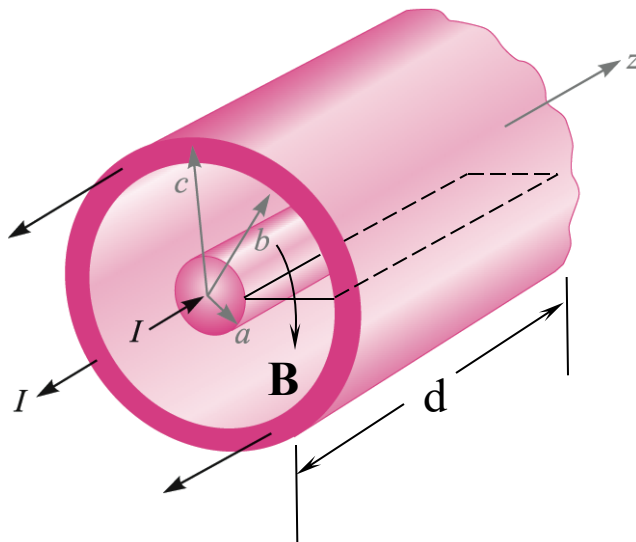
and so: $\mathbf{B} = \mu_0 \mathbf{H} = \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_{\phi}$

The magnetic flux is now the integral of $\mathbf{B}$ over the flat surface between radii a and b , and of length d along z :

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_0^d \int_a^b \frac{\mu_0 I}{2\pi\rho} \mathbf{a}_{\phi} \cdot d\rho dz \mathbf{a}_{\phi}$$

The result is: $\Phi = \frac{\mu_0 I d}{2\pi} \ln \frac{b}{a}$

The coax line thus “stores” this amount of magnetic flux in the region between conductors. This will have importance when we discuss inductance in a later lecture.



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- Scalar Magnetic Potential 标量磁位

We are already familiar with the relation between the scalar **electric potential** and electric field:

$$\mathbf{E} = -\nabla V$$

So it is tempting to **define a scalar magnetic potential such that**:

$$\mathbf{H} = -\nabla V_m$$

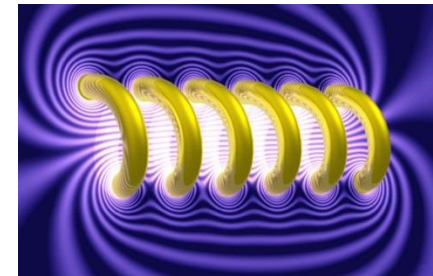
This rule must be consistent with Maxwell's equations, so therefore:

$$\nabla \times \mathbf{H} = \mathbf{J} = \nabla \times (-\nabla V_m)$$

But **the curl of the gradient of any function is identically zero!** Therefore, the scalar magnetic potential is valid only in regions where the current density is zero (such as in free space).

So we define scalar magnetic potential with a condition:

$$\mathbf{H} = -\nabla V_m \quad (\mathbf{J} = 0)$$





11: The Steady Magnetic Field

- Further Requirements on the Scalar Magnetic Potential

The other Maxwell equation involving magnetic field must also be satisfied. This is:

$$\nabla \cdot \mathbf{B} = \mu_0 \nabla \cdot \mathbf{H} = 0 \quad \text{in free space}$$

Therefore:

$$\mu_0 \nabla \cdot (-\nabla V_m) = 0$$

..and so the scalar magnetic potential satisfies Laplace's equation (again with the restriction that current density must be zero):

$$\nabla^2 V_m = 0 \quad (\mathbf{J} = 0)$$

Note: Poisson's equation for electrostatic field:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon}$$

11: The Steady Magnetic Field

- Example: Coaxial Transmission Line

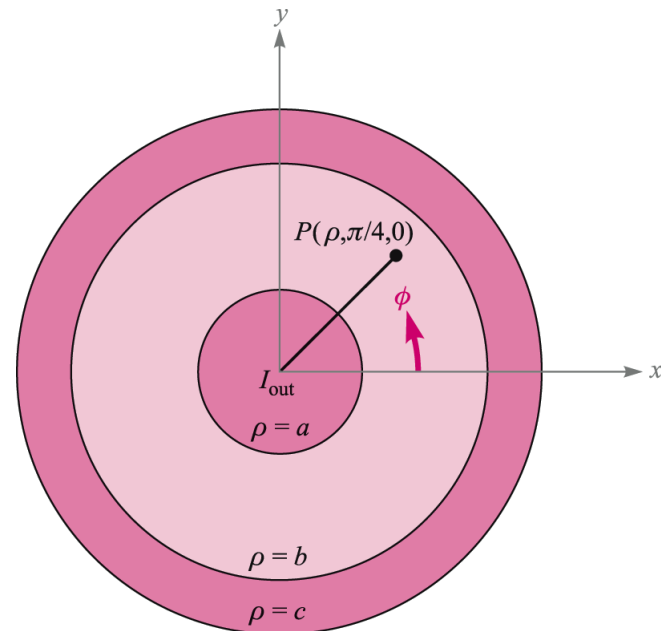
With the center conductor current flowing out of the screen, we have

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi$$

$$\text{Thus: } \frac{I}{2\pi\rho} = -\nabla V_m \Big|_\phi = -\frac{1}{\rho} \frac{\partial V_m}{\partial \phi}$$

$$\text{So we solve: } \frac{\partial V_m}{\partial \phi} = -\frac{I}{2\pi}$$

$$\text{.. and obtain: } \underline{V_m = -\frac{I}{2\pi}\phi}$$



where the integration constant has been set to zero

11: The Steady Magnetic Field

- Ambiguities in the Scalar Magnetic Potential

The scalar potential is now:

$$V_m = -\frac{I}{2\pi}\phi$$

where the potential is zero at $\phi = 0$

At point P ($\phi = \pi/4$) the potential is

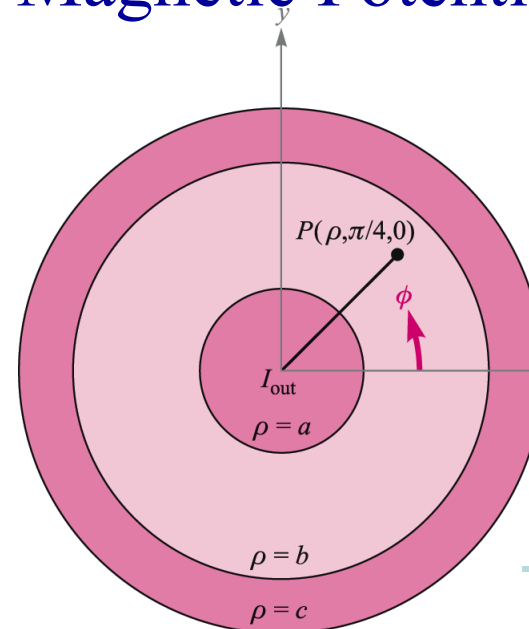
$$V_{mP}(\phi = \pi/4) = -I/8$$

But wait! As ϕ increases to $\phi = 2\pi$

we have returned to the same physical location, and the potential has a new value of $-I$.

In general, the potential at P will be **multivalued**, and will acquire a new value after each full rotation in the xy plane:

$$V_{mP} = \frac{I}{2\pi} \left(2n - \frac{1}{4}\right) \pi \quad (n = 0, \pm 1, \pm 2, \dots)$$



$$\nabla \times \mathbf{E} = 0$$

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$V_{ab} = -\int_b^a \mathbf{E} \cdot d\mathbf{L}$$

$$\nabla \times \mathbf{H} = 0 \quad (\text{wherever } \mathbf{J} = 0)$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

11: The Steady Magnetic Field

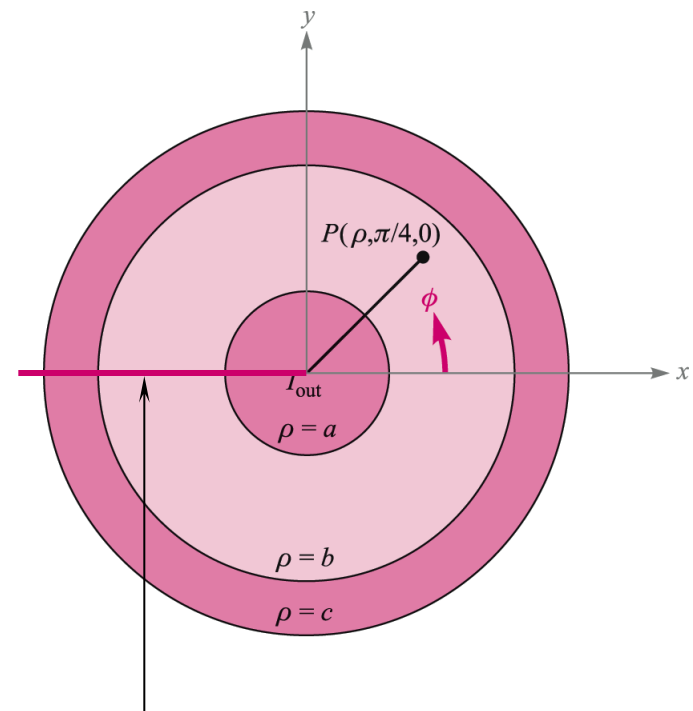
- Overcoming the Ambiguity

To remove the ambiguity, we **construct a mathematical barrier at any value of ϕ** . The angle domain cannot cross this barrier in either direction, and so the potential function is restricted to angles on either side. **In the present case we choose the barrier to lie at $\phi = \pi$ so that**

$$V_m = -\frac{I}{2\pi}\phi \quad (-\pi < \phi < \pi)$$

The potential at point P is now single-valued:

$$V_{mP} = -\frac{I}{8} \quad \left(\phi = \frac{\pi}{4}\right)$$



Barrier $\phi = \pi$



11: The Steady Magnetic Field

- Vector Magnetic Potential 矢量磁位, 磁矢位

We make use of the Maxwell equation: $\nabla \cdot \mathbf{B} = 0$

.. and the fact that the divergence of the curl of any vector field is identically zero (show this!)

$$\nabla \cdot \nabla \times \mathbf{A} = 0$$

This leads to the definition of the magnetic vector potential, $\mathbf{A}$:

$$\underline{\mathbf{B} = \nabla \times \mathbf{A}}$$

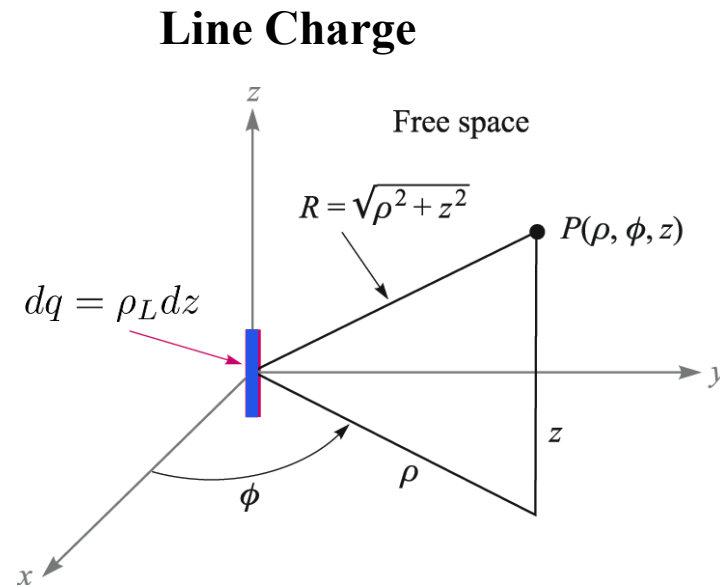
Thus: $\mathbf{H} = \frac{1}{\mu_0} \nabla \times \mathbf{A}$

and Ampere's Law becomes $\nabla \times \mathbf{H} = \mathbf{J} = \frac{1}{\mu_0} \nabla \times \nabla \times \mathbf{A}$

11: The Steady Magnetic Field

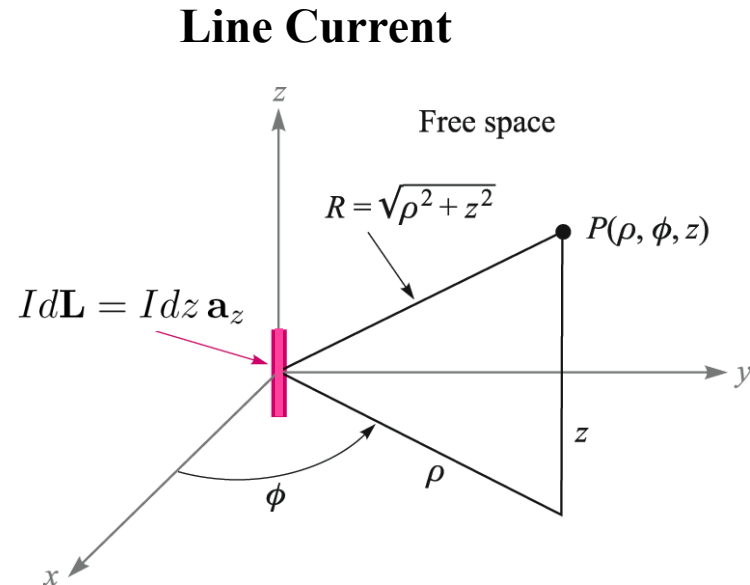
- Expressions for Potential

Consider a differential elements, shown here. On the left is a point charge represented by a differential length of line charge. On the right is a differential current element. The setups for obtaining potential are identical between the two cases.



Scalar Electrostatic Potential

$$dV = \frac{dq}{4\pi\epsilon_0 R} = \frac{\rho_L dL}{4\pi\epsilon_0 R}$$



Vector Magnetic Potential

$$d\mathbf{A} = \frac{\mu_0 Id\mathbf{L}}{4\pi R} = \frac{\mu_0 Idz \mathbf{a}_z}{4\pi R}$$

11: The Steady Magnetic Field

- General Expressions for Vector Potential

For large scale charge or current distributions, we would sum the differential contributions by integrating over the charge or current, thus:

$$\mathbf{A} = \oint \frac{\mu_0 I d\mathbf{L}}{4\pi R}$$

The closed path integral is taken because the current must close on itself to form a complete circuit.

For surface or volume current distributions, we would have, respectively:

$$\mathbf{A} = \int_S \frac{\mu_0 \mathbf{K} dS}{4\pi R}$$

or

$$\mathbf{A} = \int_{\text{vol}} \frac{\mu_0 \mathbf{J} dv}{4\pi R}$$

in the same manner that we used for scalar electric potential.

11: The Steady Magnetic Field

- Example

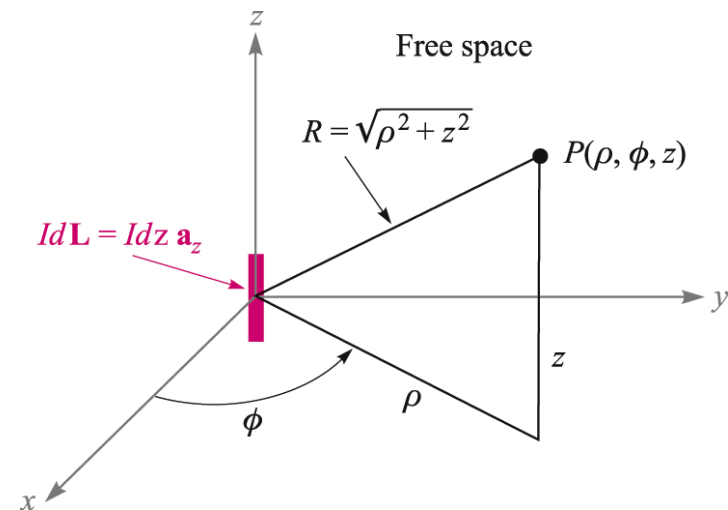
We continue with the differential current element as shown here:

In this case

$$d\mathbf{A} = \frac{\mu_0 I d\mathbf{L}}{4\pi R}$$

becomes at point P:

$$d\mathbf{A} = \frac{\mu_0 I dz \mathbf{a}_z}{4\pi\sqrt{\rho^2 + z^2}}$$



Now, the curl is taken in cylindrical coordinates:

$$d\mathbf{H} = \frac{1}{\mu_0} \nabla \times d\mathbf{A} = \frac{1}{\mu_0} \left(-\frac{\partial dA_z}{\partial \rho} \right) \mathbf{a}_\phi = \frac{I dz}{4\pi} \frac{\rho}{(\rho^2 + z^2)^{3/2}} \mathbf{a}_\phi$$

This is the same result as found using the Biot-Savart Law (as it should be)

11: The Steady Magnetic Field

- Equation for the Vector Potential

We start with: $\nabla \times \mathbf{H} = \mathbf{J} = \frac{1}{\mu_0} \nabla \times \nabla \times \mathbf{A}$

Then, introduce a vector identity that defines **the vector Laplacian**:

$$\nabla^2 \mathbf{A} \equiv \nabla^2 A_x \mathbf{a}_x + \nabla^2 A_y \mathbf{a}_y + \nabla^2 A_z \mathbf{a}_z$$

$$\nabla^2 \mathbf{A} \equiv \nabla(\nabla \cdot \mathbf{A}) - \nabla \times \nabla \times \mathbf{A}$$

Using a (lengthy) procedure (see Sec. 7.7) it can be proven that

$$\underline{\nabla \cdot \mathbf{A} = 0}$$

We are therefore left with

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$



11: The Steady Magnetic Field

- Equation for the Vector Potential

We now have

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

In rectangular coordinates:

$$\nabla^2 \mathbf{A} = \nabla^2 A_x \mathbf{a}_x + \nabla^2 A_y \mathbf{a}_y + \nabla^2 A_z \mathbf{a}_z \quad (\text{not so simple in the other coordinate systems})$$

The equation separates to give:

$$\nabla^2 A_x = -\mu_0 J_x$$

$$\nabla^2 A_y = -\mu_0 J_y$$

$$\nabla^2 A_z = -\mu_0 J_z$$

This indicates that **the direction of $\mathbf{A}$ will be the same as that of the current to which it is associated.** Also, we get three scalar Poisson's equations.

HW-7



2. A filamentary conductor is formed into an equilateral triangle with sides of length l , carrying current I . Find the magnetic field intensity $\mathbf{H}$ at the center of the triangle.
3. Two circular current loops are centered on the z axis at $z = \pm h$. Each loop has radius a and carries current I in the $\mathbf{a}_\phi$ direction. Find $\mathbf{H}$ on the z axis over the range $-h < z < h$.
4. Let $\mathbf{A} = (3y - z)\mathbf{a}_x + 2xz\mathbf{a}_y$ Wb/m in a certain region of free space.
(a) Show that $\nabla \cdot \mathbf{A} = 0$. (b) At $P(2, -1, 3)$, find $\mathbf{A}$, $\mathbf{B}$, $\mathbf{H}$, and $\mathbf{J}$.
5. Use an expansion in rectangular coordinates to show that the curl of the gradient of any scalar field G is identically equal to zero.